University of Louisville

Experimental Psychology Doctoral Program

Semester Student Activity Report

INSTRUCTIONS	
1	Complete form
2	Meet with mentor to review semester activities and goals
3	Save to PDF
4	Sign and have mentor sign report electronically
5	Upload to Box folder before deadline

STUDENT NAME	Siraj Lyons
MENTOR NAME	Brendan Depue
ACADEMIC SEMESTER/YEAR	Spring 2026
YEAR IN PROGRAM	4th
DATE DEGREE EXPECTED	May 2027

MILESTONES

	Date	Outcome
MS PORTFOLIO	Summer 2024	☒Pass ☐ Fail
PRELIMINARY EXAM PROPOSAL	May 2025	☒Pass ☐ Fail
PRELIMINARY EXAM DEFENSE	August 2025	☒Pass ☐ Fail
DISSERTATION PROPOSAL	March 2026	☒Pass ☐ Fail
DISSERTATION DEFENSE	N/A	☐Pass ☐ Fail

FUNDING SOURCE THIS SEMESTER

☒GTA
☐GRA
☐ Fellowship
☐ Grant
☐Other _____

STUDENT LEARNING OUTCOMES FOR UNIVERSITY REPORTS

Do you have current Human Subjects Ethics Training (CITI and HIPAA) approval?	☒Yes ☐ No	In the past year, did you attend at least 75% of Department Colloquia (e.g., Grawemeyer talks)?	☒Yes ☐ No
How many times have you presented your research at RAPDS while in the program?	3	How many regional or national conferences have you attended while in the program?	2
How many oral or poster presentations have you given at scientific conferences while in the program?	3	How many professional development sessions have you attended while in the program (outside of RAPDS)?	1
Have you presented at an applied conference, meeting, workshop, or other setting (including a written outlet), or engaged in professional service within the department, university, or community, while in the program?	☒Yes ☐ No	How many training workshops, webinars, research talks, or committee meetings related to diversity and/or inclusion have you attended while in the program (including RAPDS)?	8
How many manuscripts have you submitted for publication while in the program?	6		

COURSE REQUIREMENTS

Courses taken in Years 1-3 of the Graduate Program (leave semesters you have not completed blank)

SUMMER YEAR 1				FALL YEAR 1				SPRING YEAR 1			
COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE
Independent Study - Reading	604	3		Advanced Statistics I	610	3	A+	Advanced Statistics II	611	3	A+
				Independent Research	605	2	A	Independent Research	605	2	A
				Seminar: Special Populations	609	3	A	Cognitive Processes	621	3	A
				Research in Experimental Psychology	602	1	A	Research in Experimental Psychology	602		A
				Research in Experimental Psychology	602	1	A				

SUMMER YEAR 2				FALL YEAR 2				SPRING YEAR 2			
COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE
Independent Research	605	6	A	Independent Research	605	2	A	Independent Research	605	2	A
				Seminar: Academic Writing	609	3	A	Seminar: Neuroscience of Memory	609	3	A
				Behavioral Neuroscience		3	A	Decision Sciences	638	3	A
				Research in Experimental Psychology	602	1	A	Research In Experimental Psychology	602	1	A

SUMMER YEAR 3				FALL YEAR 3				SPRING YEAR 3			
COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE
Independent Research	605	6	A	Independent Research	605	8	N/A				
				Research in Experimental Psychology	602	1	N/A				

List Other Courses Taken Past Year 3 (If Applicable)

COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE	COURSE TITLE	COURSE #	CREDIT HRS	GRADE

Will you be enrolled in Doctoral Candidacy next semester (fall/spring/summer)? ☒Yes ☐ No

RESEARCH ACTIVITIES

PUBLICATIONS **Number of books, book chapters, or articles in peer reviewed professional/scientific journals that you were an author or co-author on** *Include in-press publications; Under review or submitted should NOT be counted here* *List citations of all your UofL publications below. Put an *asterisk by those published this semester*	4

Foley, F., **Lyons, S. J.**, Cook, O., & Depue, B. E. (2026). Investigating the Functional Neural Network Dynamics of Cognitive-Affective Action Planning during Threat Processing. *Frontiers in Psychology*.

Lawrence, J. M., Foote, I. F., Breunig, S., Schaffer, L. S., **Lyons, S.**, Abramowitz, S., Levin, M. G., Damrauer, S. M., Mallard, T. T., Grotzinger, A. D. (2026). Shared genetic liability across systems of psychiatric and physical illness. *Nature Communications*. https://doi.org/10.1038/s41467-026-69218-1

Lyons, S., Beck, I., & Depue, B. E. (2025). Depression is marked by differences in structural covariance between deep brain nuclei and sensorimotor cortex. *NeuroImage*, 310. https://doi.org/10.1016/j.neuroimage.2025.121127

Lyons, S. & Depue, B. E. (2023). Not all bad decisions are alike: Approach and avoidant bad decisions are associated with distinct network organization. *Frontiers in Neuroscience*, *17*, Article number 1249008. https://doi.org/10.3389/fnins.2023.1249008

PUBLICATIONS SUBMITTED, UNDER REVIEW **Number of publications under review (books, book chapters, or articles in peer reviewed professional/scientific journals that you were an author or co-author on)** *List citations of all under-review publications below. Put an *asterisk by those submitted this semester*	2

Lyons, S. J., Foley, J., Cook, O., & Depue, B. E. (*Under Review; R1*). Bioinformed idiographic symptom networks to identify neurobiological predictors of affective-state transitions in bipolar disorder. *Imaging Neuroscience*

Lyons, S. J., & Depue, B. E. (*Initial Review*). Sensorimotor inference as an adjunctive model of bipolar disorder. *Neuroscience and Biobehavioral Reviews*

OTHER PAPERS IN PROGRESS **Number of publications currently being written, data collection completed (books, book chapters, or articles in peer reviewed professional/scientific journals that you are an author or co-author)** *List working title and authors below. Put an *asterisk by those you actively worked on this semester*	2

Lyons, S. J., Saki, J., Reid, A., Christman, S. D., & Jasper, J. D. (*In Prep*). Effects of leftward unilateral gaze on Iowa Gambling Task performance across the hypomanic personality spectrum depends on handedness. *Target Journal: Laterality*.

Lyons, S. J., Foley, J. F., Gruner, S.*, & Fang, L. (*In Prep*). Personality clusters predicting attention bias toward monetary images: A data-driven approach. [Working Title].

RESEARCH PRESENTATIONS **Number of workshops, oral presentations, or poster presentations at professional meetings you were an author or co-author on** *Include accepted presentations* *List citations of all your UofL presentations below. Put an *asterisk by those presented this semester*	3 Conferences; 4 Posters

Lyons, S., Beck, I., & Depue, B. (March – April, 2025). Structural co-dependence between deep brain nuclei and sensorimotor cortex differs between people with depression and unaffected individuals.

Lyons, S., Irvin, Z., & Depue, B. (November 2023). Frontoparietal network stability predicts attentional control and cognitive flexibility. Society for Neuroscience.

Lyons, S. & Depue, B. (March 2023). Behavioral approach is linked to increased neural communica5on between frontostriatal and insular regions during risky decision-making. Cognitive Neuroscience Society.

Cook, O., Lyons, S., & Depue, B. (March 2023). Cognitive and motor neural network interation during threat response. Cognitive Neuroscience Society.

GRANTS **Have you submitted or received any grants?** *If yes, please list citation below: authors, title of grant, name of funding source, amount requested, and if grant was approved/money awarded. Put an *asterisk by those submitted this semester*	0

N/A

OTHER RESEARCH ACTIVITY
What other research activity did you actively work on this semester, that does not fall into the above categories (e.g., data collected, literature review, studies planned, etc.)?
List details below.

Preparing a grant application to the Wright Foundation to study depression and hypomania/mania symptoms across top-down and bottom-up cognitive domains. Three studies are planned.

- Cued-switching task (top-down processing)
- Stimulus detection task (bottom-up processing)
- Mobile EEG pilot to track nonlinear neural trajectories (technology to be acquired if grant is approved)
- Developed an EEG-processing pipeline to establish feasibility of mobile EEG pilot for grant

Large-archival analysis of structural MRI data to derive gray matter networks, their relationship with brainstem nuclei, and individual differences

- Primarily structured as an advisory project where I mentor two undergraduate research assistants through MRI processing

Obtained access to MRI consortium data possessing 3,000 scans to fit finite mixture models on MRI to identify subgroups based upon cortical resting-state entropy and their clinical associations.

Began extramural collaboration on project developing novel MRI analytical techniques.

TEACHING ACTIVITIES

TEACHING ASSISTANTSHIPS		
SEMESTER/YEAR	**COURSE**	**PROFESSOR**
Fall 2024	PSYC 301	Maria Kondaurova
Spring 2025	PSYC 382	Andrew Lynn
Fall 2025	PSYC 305	Kamryn Mattingly
Fall 2025	PSYC 305 (Distance)	Brendan Depue
Spring 2026	PSYC 301	Maria Kondaurova

COURSES TAUGHT	
SEMESTER/YEAR	**COURSE**

GUEST LECTURES		
SEMESTER/YEAR	**COURSE**	**PROFESSOR**
Spring 2025	PSYC 382 (*Emotion*)	Andrew Lynn

SERVICE ACTIVITIES

Please list any service activities with the department, university, community, etc. (committees, SGA offices held, community outreach based on your professional role)
Peer Mentorship GNAS Representative (Chair of Academic and Research Committee) PLUS Committee (Assessments Subcommittee)

PREVIOUS RESEARCH GOALS

What research goals did you previously list for this past semester? (Copy/paste all of the goals you submitted last semester)
Propose dissertation. Submit an opinion/perspective article currently in preparation. Start collecting data behavioral project investigating signal detection theory. Start processing behavioral data collected for a joint project between NILCAMP and CABIN (Northern Mich. Univ.). Also, begin writing the manuscript associated with this data. Start Experiment 1 in Dissertation.

STUDENT COMMENTS ON PROGRESS

Provide your own self-evaluation of your performance regarding your research goals this past semester. Reflect on your accomplishments. Reflect on goals you did not meet. Are there any specific areas or concerns that you would like to discuss with your mentor or program director?

Goals met: Propose dissertation, submitted opinion article to Neurosicence and Biobehavioral Reviews, began processing behavioral data for joint behavioral project, began writing manuscript associated with the joint behavioral project.

Goals thwarted: Began dissertation 1 in dissertation.

Evaluation: I accomplished the majority of my goals during this semester and believe that I'm on track to graduate in May 2027. My only thwarted goal was to begin the first experiment of my dissertation. This was due to delays in my dissertation proposal manifested by changing the scope of the project at the last-minute. This occurred after I learned about the difficulties of obtaining archival data from the ABCD project. Consequently, I proposed in late March and finished the Spring semester in a state of analysis preparation. I do anticipate to have analyses running starting in early May.

In reflection, I believe I am developing adequately to becoming an independent researcher. I submitted two short review/opinion articles which were seeded by my preliminary exam. One is currently past the first-round of revisions at *Imaging Neuroscience* and outlines both the feasibility and importance of dense longitudinal neuroimaging to find temporal predictors of affective episodes using neurobiological measures. This paper was an extension of one minor theme which emerged from my preliminary exam. The other is under review at *Neuroscience and Biobehavioral Reviews* which more closely aligns to my primary topic covered in the preliminary exam. I revised the themes based on feedback from my committee and presented a theoretical case to intentionally/explicitly model sensory and motoric processes which co-occur with reward/goal-directed behavior in bipolar disorder.

I am pleased that I submitted two first-author papers during this academic year, which were predominately theory driven. However, I am aware that empirical articles ought to be my focus as these carry more weight for hiring committees.

In my second-author publication published this semester, I learned an applied support-vector machine classification to identify MRI-predictors of threat- and non-threat conditions. Moreover, during my recent endeavor drafting a grant, I refreshed my familiarity with EEG and learned how to perform spectral analyses.

My time analyzing EEG data will eventually be transferred to the iEEG project which remains in development. Tasks have been designed for the project, however it appears likely that my initial work will be limited to resting-state processing.

NEW RESEARCH GOALS

What research goals do you have for next semester? (Be specific. After writing these, discuss with your mentor.)
-Apply for intramural research grant (Wright Foundation) to fund behavioral investigations of mood symptoms and an investigatory project studying mobile neural trajectories across mood symptoms -Produce draft for NIH-affiliated grant to be used when seeking postdoc positions. This grant will largely align with work to be submitted to the Wright Foundation with modifications to focus on topics highlighted in my under-review paper to *Neuroscience and Biobehavioral Reviews*. -Attend a scientific conference to present work on entropy in depression OR preliminary outcomes of my EEG analyses -Submit two empirical articles

The content of this form is accurate, and I agree with the stated goals for next semester and understand that I am responsible for actively working towards these goals and staying on track.

STUDENT SIGNATURE		**DATE**	

I have reviewed the content of this form and agree to mentor the student as they work towards their goals for the next semester.

MENTOR SIGNATURE		**DATE**	**4/28/2026**